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Aster Models for Life History Analysis
Author(s): Charles J. Geyer, Stuart Wagenius and Ruth G. Shaw
Source: *Biometrika*, Vol. 94, No. 2 (Jun., 2007), pp. 415-426
Published by: Oxford University Press on behalf of Biometrika Trust
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Biometrika (2007), **94**, 2, *pp.* 415–426
doi:10.1093/biomet/asm030
Advance Access publication 14 May 2007
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Printed in Great Britain

Aster models for life history analysis

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SUMMARY

We present a new class of statistical models, designed for life history analysis of plants and animals, that allow joint analysis of data on survival and reproduction over multiple years, allow for variables having different probability distributions, and correctly account for the dependence of variables on earlier variables. We illustrate their utility with an analysis of data taken from an experimental study of *Echinacea angustifolia* sampled from remnant prairie populations in western Minnesota. These models generalize both generalized linear models and survival analysis. The joint distribution is factorized as a product of conditional distributions, each an exponential family with the conditioning variable being the sample size of the conditional distribution. The model may be heterogeneous, each conditional distribution being from a different exponential family. We show that the joint distribution is from a flat exponential family and derive its canonical parameters, Fisher information and other properties. These models are implemented in an R package 'aster' available from the Comprehensive R Archive Network, CRAN.

Some key words: Conditional exponential family; Flat exponential family; Generalized linear model; Graphical model; Maximum likelihood.

1. INTRODUCTION

This article introduces a class of statistical models we call 'aster models'. They were invented for life history analysis of plants and animals and are best introduced by an example about perennial plants observed over several years. For each individual planted, at each census, we record whether or not it is alive, whether or not it flowers, and its number of flower heads. These data are complicated, especially when recorded for several years, but simple conditional models may suffice. We consider mortality status, dead or

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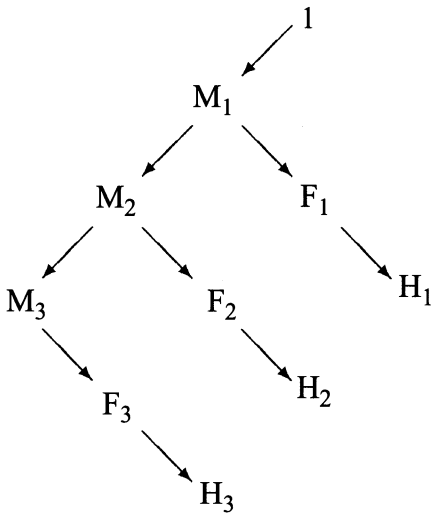


Fig. 1. Graph for *Echinacea* aster data. Arrows go from parent nodes to child nodes. Nodes are labelled by their associated variables. The only root node is associated with the constant variable 1. M_j is the mortality status in year $2001 + j$. F_j is the flowering status in year $2001 + j$. H_j is the flower head count in year $2001 + j$. The M_j and F_j are Bernoulli conditional on their parent variables being one, and zero otherwise. The H_j are zero-truncated Poisson conditional on their parent variables being one, and zero otherwise.

alive, to be Bernoulli given the preceding mortality status. Similarly, flowering status given mortality status is also Bernoulli. Given flowering, the number of flower heads may have a zero-truncated Poisson distribution (Martin et al., 2005). Figure 1 shows the graphical model for a single individual.

This aster model generalizes both discrete time Cox regression (Cox, 1972; Breslow, 1972, 1974) and generalized linear models (McCullagh & Nelder, 1989). Aster models apply to any similar conditional modelling. We could, for example, add other variables, such as seed count modelled conditional on flower head count.

A simultaneous analysis that models the joint distribution of all the variables in a life history analysis can answer questions that cannot be addressed through separate analyses of each variable conditional on the others.

Joint analysis also deals with structural zeros in the data; for example, a dead individual remains dead and cannot flower, so in Fig. 1 any arrow that leads from a variable that is zero to another variable implies that the other variable must also be zero. Such zeros present intractable missing data problems in separate analyses of individual variables. Aster models have no problem with structural zeros; likelihood inference automatically handles them correctly.

Aster models are simple graphical models (Lauritzen, 1996, §3·2·3) in which the joint density is a product of conditionals as in equation (1) below. No knowledge of graphical model theory is needed to understand aster models. One innovative aspect of aster models is the interplay between two parameterizations described in §§2·2 and 2·3 below. The

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'conditional canonical parameterization' arises when each conditional distribution in the product is an exponential family and we use the canonical parameterization for each. The 'unconditional canonical parameterization' arises from observing that the joint model is a full flat exponential family (Barndorff-Nielsen, 1978, Ch. 8) and using the canonical parameters for that family, defined by equation (5) below.

2. ASTER MODELS

2·1. *Factorization and graphical model*

Variables in an aster model are denoted by X_j, where j runs over the nodes of a graph. A general aster model is a chain graph model (Lauritzen, 1996, pp. 7, 53) having both arrows, corresponding to directed edges, and lines, corresponding to undirected edges. Figure 1 is special, having only arrows. Arrows go from parent to child, and lines between neighbours. Nodes that are not children are called root nodes. Those that are not parents are called terminal nodes.

Let F and J denote root and nonroot nodes. Aster models have very special chain graph structure determined by a partition $\mathcal{G}$ of J and a function $p : \mathcal{G} \to J \cup F$. For each $G \in \mathcal{G}$ there is an arrow from $p(G)$ to each element of G and a line between each pair of elements of G. For any set S, let X_S denote the vector whose components are X_j, $j \in S$. The graph determines a factorization

$$\mathrm{pr}(X_J|X_F) = \prod_{G\in\mathcal{G}} \mathrm{pr}\{X_G|X_{p(G)}\}; \tag{1}$$

compare with equation (3.23) in Lauritzen (1996).

Elements of $\mathcal{G}$ are called chain components because they are connectivity components of the chain graph (Lauritzen, 1996, pp. 6–7). Since Fig. 1 has no undirected edge, each node is a chain component by itself. Allowing nontrivial chain components allows the elements of X_G to be conditionally dependent given $X_{p(G)}$ with merely notational changes to the theory. In our example in §5 the graph consists of many copies of Fig. 1, one for each individual plant. Individuals have no explicit representation. For any set S, let $p^{-1}(S)$ denote the set of G such that $p(G) \in S$. Then each subgraph consisting of one $G \in p^{-1}(F)$, its descendants, children, children of children, etc., and arrows and lines connecting them, corresponds to one individual. If we make each such G have a distinct root element $p(G)$, then the set of descendants of each root node corresponds to one individual. Although all individuals in our example have the same subgraph, this is not required.

2·2. *Conditional exponential families*

We take each of the conditional distributions in (1) to be an exponential family having canonical statistic X_G that is the sum of $X_{p(G)}$ independent and identically distributed random vectors, possibly a different such family for each G. Conditionally, $X_{p(G)} = 0$ implies that $X_G = 0_G$ almost surely. If $j \notin p(G)$ for any G, then the values of X_j are unrestricted. If the distribution of X_G given $X_{p(G)}$ is infinitely divisible, such as Poisson or normal, for each $G \in p^{-1}(\{j\})$, then X_j must be nonnegative and real-valued. Otherwise, X_j must be nonnegative and integer-valued.

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The loglikelihood for the whole family has the form

$$\sum_{G\in\mathcal{G}}\left\{\sum_{j\in G} X_j\theta_j - X_{p(G)}\psi_G(\theta_G)\right\} = \sum_{j\in J} X_j\theta_j - \sum_{G\in\mathcal{G}} X_{p(G)}\psi_G(\theta_G), \tag{2}$$

where θ_G is the canonical parameter vector for the Gth conditional family, having components θ_j, $j \in G$, and ψ_G is the cumulant function for that family (Barndorff-Nielsen, 1978, pp. 105, 139, 150) that satisfies

$$E_{\theta_G}(X_G|X_{p(G)}) = X_{p(G)}\nabla\psi_G(\theta_G) \tag{3}$$

$$\mathrm{var}_{\theta_G}(X_G|X_{p(G)}) = X_{p(G)}\nabla^2\psi_G(\theta_G), \tag{4}$$

where $\mathrm{var}_\theta(X)$ is the variance-covariance matrix of X and $\nabla^2\psi(\theta)$ is the matrix of second partial derivatives of ψ (Barndorff-Nielsen, 1978, p. 150).

2·3. *Unconditional exponential families*

Collecting terms with the same X_j in (2), we obtain

$$\sum_{j\in J} X_j\left\{\theta_j - \sum_{G\in p^{-1}(\{j\})}\psi_G(\theta_G)\right\} - \sum_{G\in p^{-1}(F)} X_{p(G)}\psi_G(\theta_G)$$

and see that

$$\varphi_j = \theta_j - \sum_{G\in p^{-1}(\{j\})}\psi_G(\theta_G), \qquad j \in J, \tag{5}$$

are the canonical parameters of an unconditional exponential family with canonical statistics X_j. We now write X instead of X_J, φ instead of φ_J, and so forth, and let $\langle X, \varphi\rangle$ denote the inner product $\sum_j X_j\varphi_j$. Then we can write the loglikelihood of this unconditional family as

$$l(\varphi) = \langle X, \varphi\rangle - \psi(\varphi), \tag{6}$$

where the cumulant function of this family is

$$\psi(\varphi) = \sum_{G\in p^{-1}(F)} X_{p(G)}\psi_G(\theta_G). \tag{7}$$

All of the $X_{p(G)}$ in (7) are at root nodes, and hence are nonrandom, so that ψ is a deterministic function. Also, the right-hand side of (7) is a function of φ by the logic of exponential families (Barndorff-Nielsen, 1978, pp. 105 ff.).

The system of equations (5) can be solved for the θ_j in terms of the φ_j in one pass through the equations in any order that finds θ_j for children before parents. Thus (5) determines an invertible change of parameter.

2·4. *Canonical affine models*

One of the desirable aspects of exponential family canonical parameter affine models defined by reparameterization of the form

$$\varphi = a + M\beta, \tag{8}$$

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where a is a known vector, called the origin, and M is a known matrix, called the model matrix, is that, because $\langle X, M\beta \rangle = \langle M^{\mathrm{T}} X, \beta \rangle$, the result is a new exponential family with canonical statistic

$$Y = M^{\mathrm{T}} X \tag{9}$$

and canonical parameter β. The dimension of this new family will be the dimension of β, if M has full rank.

As is well known (Barndorff-Nielsen, 1978, p. 111), the canonical statistic of an exponential family is minimal sufficient. Since we have both conditional and unconditional families in play, we stress that this well-known result is about unconditional families. A dimension reduction to a low-dimensional sufficient statistic like (9) does not occur when the conditional canonical parameters θ are modelled affinely, and this suggests that affine models for the unconditional parameterization may be scientifically more interesting despite their more complicated structure.

2·5. *Mean-value parameters*

Conditional mean values

$$\xi_j = E_{\theta_G}(X_j | X_{p(G)}) = X_{p(G)} \frac{\partial \psi_G(\theta_G)}{\partial \theta_j}, \qquad j \in G, \tag{10}$$

are not parameters because they contain random data $X_{p(G)}$, although they play the role of mean-value parameters when we condition on $X_{p(G)}$, treating it as constant. By standard exponential family theory (Barndorff-Nielsen, 1978, p. 121), $\nabla \psi_G$ is an invertible change of parameter.

Unconditional mean-value parameters are the unconditional expectations

$$\tau = E_\varphi(X) = \nabla \psi(\varphi). \tag{11}$$

By standard theory, $\nabla \psi : \varphi \mapsto \tau$ is an invertible change of parameter. The unconditional expectation in (11) can be calculated using the iterated expectation theorem

$$E_\varphi(X_j) = E_\varphi(X_{p(G)}) \frac{\partial \psi_G(\theta_G)}{\partial \theta_j}, \qquad j \in G, \tag{12}$$

where θ is determined from φ by solving (5). The system of equations (12) can produce the τ_j in one pass through the equations in any order that finds τ_j for parents before children.

3. Likelihood inference

3·1. *Conditional models*

The score $\nabla l(\theta)$ for conditional canonical parameters is particularly simple, having components

$$\frac{\partial l(\theta)}{\partial \theta_j} = X_j - X_{p(G)} \frac{\partial \psi_G(\theta_G)}{\partial \theta_j}, \qquad j \in G, \tag{13}$$

and, if these parameters are modelled affinely as in (8) but with φ replaced by θ, then

$$\nabla l(\beta) = \nabla l(\theta) M. \tag{14}$$

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The observed Fisher information matrix for θ, which is the matrix $-\nabla^2 l(\theta)$, is block diagonal with

$$-\frac{\partial^2 l(\theta)}{\partial\theta_i \partial\theta_j} = X_{p(G)} \frac{\partial^2 \psi_G(\theta_G)}{\partial\theta_i \partial\theta_j}, \qquad i, j \in G, \tag{15}$$

the only nonzero entries. The expected Fisher information matrix for θ is the unconditional expectation of the observed Fisher information matrix, calculated using (15) and (12). If $I(\theta)$ denotes either the observed or the expected Fisher information matrix for θ and similarly for other parameters, then

$$I(\beta) = M^{\mathrm{T}} I(\theta) M. \tag{16}$$

3·2. *Unconditional models*

The score $\nabla l(\varphi)$ for unconditional canonical parameters is, as in every unconditional exponential family, 'observed minus expected':

$$\frac{\partial l(\varphi)}{\partial \varphi_j} = X_j - E_\varphi(X_j),$$

the unconditional expectation on the right-hand side being evaluated by using (12). If these parameters are modelled affinely as in (8), then we have (14) with θ replaced by φ. Note that (13) is not 'observed minus conditionally expected' if considered as a vector equation because the conditioning would differ amongst components.

Second derivatives with respect to unconditional canonical parameters of an exponential family are nonrandom, and hence observed and expected Fisher information matrices $I(\varphi)$ are equal, given by either of the expressions $\nabla^2 \psi(\varphi)$ and $\mathrm{var}_\varphi(X)$. Fix θ and φ related by (5). For $i, j \in G$, the iterated covariance formula gives

$$\mathrm{cov}_\varphi(X_i, X_j) = \frac{\partial^2 \psi_G(\theta_G)}{\partial\theta_i \partial\theta_j} E_\varphi\big(X_{p(G)}\big) + \frac{\partial \psi_G(\theta_G)}{\partial\theta_i} \frac{\partial \psi_G(\theta_G)}{\partial\theta_j} \mathrm{var}_\varphi\big(X_{p(G)}\big). \tag{17}$$

Otherwise we may assume that $j \in G$ and i is not a descendant of j so that $\mathrm{cov}_\varphi(X_i, X_j | X_{p(G)}) = 0$ because X_j is conditionally independent given $X_{p(G)}$ of all variables except itself and its descendants. Then the iterated covariance formula gives

$$\mathrm{cov}_\varphi(X_i, X_j) = \frac{\partial \psi_G(\theta_G)}{\partial\theta_j} \mathrm{cov}_\varphi\big(X_i, X_{p(G)}\big). \tag{18}$$

Expectations having been calculated using (12), variances, the case $i = j$ in (17), are calculated in one pass through (17) in any order that deals with parents before children. Then another pass using (17) and (18) gives covariances. The information matrix for β is given by (16) with θ replaced by φ.

3·3. *Prediction*

By 'prediction', we only mean evaluation of a function of estimated parameters, what the `predict` function in R does for generalized linear models. In aster models we have five different parameterizations of interest, $\beta, \theta, \varphi, \xi$ and τ. The Fisher information matrix for β, already described, handles predictions of β, so this section is about 'predicting' the remaining four. One often predicts for new individuals having different covariate values

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from the observed individuals. Then the model matrix $\tilde{M}$ used for the prediction is different from that used for calculating $\hat{\beta}$ and the Fisher information matrix $I(\hat{\beta})$, either observed or expected.

Let η be the affine predictor, i. e. $\eta = \theta$ for conditional models and $\eta = \varphi$ for unconditional models, let ζ be any one of θ, φ, ξ and τ, and let $f : \eta \mapsto \zeta$. Suppose we wish to predict

$$g(\beta) = h(\zeta) = h\{f(\tilde{M}\beta)\}. \tag{19}$$

Then, by the chain rule, (19) has derivative

$$\nabla g(\beta) = \nabla h(\zeta) \nabla f(\eta) \tilde{M}, \tag{20}$$

and, by the 'usual' asymptotics of maximum likelihood and the delta method, the asymptotic distribution of the prediction $h(\hat{\zeta}) = g(\hat{\beta})$ is

$$N\big[g(\beta), \{\nabla g(\hat{\beta})\} I(\hat{\beta})^{-1} \{\nabla g(\hat{\beta})\}^{\mathrm{T}}\big],$$

where $\nabla g(\hat{\beta})$ is given by (20) with $\hat{\eta} = \tilde{M}\hat{\beta}$ plugged in for η and $\hat{\zeta} = f(\hat{\eta})$ plugged in for ζ. We write 'predictions' in this complicated form to separate the parts of the specification, the functions h and ∇h and the model matrix $\tilde{M}$, that change from application to application from the part ∇f that does not change and can be dealt with by computer; see Appendix A of the technical report at http://www.stat.umn.edu/geyer/aster for details.

To predict mean-value parameters one must specify new 'response' data X_j as well as new 'covariate' data in $\tilde{M}$. Unconditional mean-value parameters τ depend on X_j, $j \in F$, whereas conditional mean-value parameters ξ depend on X_j, $j \in J \cup F$. It is often interesting, however, to predict ξ for hypothetical individuals with $X_j = 1$ for all j, thus obtaining conditional mean values per unit sample size.

4. Software

We have released an R (R Development Core Team, 2004) package `aster` that fits, tests, predicts and simulates aster models. It uses the R formula mini-language, originally developed for GENSTAT and S (Wilkinson & Rogers, 1973; Chambers & Hastie, 1992) so that model fitting is very like that for linear or generalized linear models. The R function `summary.aster` provides regression coefficients with standard errors, z statistics and p-values; `anova.aster` provides likelihood ratio tests for model comparison; `predict.aster` provides the predictions with standard errors described in §3·3; `raster` generates random aster model data for simulation studies or parametric bootstrap calculations. The package is available from CRAN (http://www.cran.r-project.org) and is open source.

The current version of the package limits the general model described in this article in several ways. In predictions, only linear h are allowed in (19), but this can be worked around. For general h, observe that $h(\hat{\zeta})$ and $A^{\mathrm{T}}\hat{\zeta}$, where $A = \nabla h(\zeta)$, have the same standard errors. Thus, obtain $h(\hat{\zeta})$ by one call to `predict.aster` and the standard errors for $A^{\mathrm{T}}\hat{\zeta}$, where $A = \nabla h(\hat{\zeta})$ by a second call. In models, the only conditional families currently implemented are Bernoulli, Poisson, k-truncated Poisson, negative binomial and k-truncated negative binomial, but adding another one-parameter exponential family would require only implementation of its ψ, ψ' and ψ'' functions and its random variate generator. Multiparameter conditional families, chain components, are not yet implemented. Allowing

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Table 1. *Tests for model comparison. The model formulae are given above and the analysis of deviance below; deviance is double the log likelihood ratio*

```
Model 1: resp ~ varb + level:(nsloc + ewloc)
Model 2: resp ~ varb + level:(nsloc + ewloc) + hdct * pop - pop
Model 3: resp ~ varb + level:(nsloc + ewloc) + hdct * pop
Model 4: resp ~ varb + level:(nsloc + ewloc) + level * pop
```

Model Number	Model d. f.	Model Deviance	Test d. f.	Test Deviance	Test *p*-value
1	15	2728·72			
2	21	2712·54	6	16·18	0·013
3	27	2684·86	6	27·67	0·00011
4	33	2674·70	6	10·17	0·12

d. f., degrees of freedom.

terminal nodes that are two-parameter normal or allowing child nodes that are multinomial given a common parent would require more extensive changes to the package.

5. EXAMPLE

Data were collected on 570 individuals of *Echinacea angustifolia*, each having the data structure shown in Fig. 1. These plants were sampled as seeds from seven remnant populations that are surviving fragments of the tall-grass prairie that a century ago covered western Minnesota and other parts of the Great Plains of North America. The plants were experimentally randomized at the time of planting into a field within 6.5 km of all populations of origin. The dataset contains three predictor variables: `ewloc` and `nsloc` give east-west and north-south positions of individuals and `pop` gives their remnant population of origin. To use the R formula mini-language we need to create more variables: `resp` is a vector comprising the response variables, the M_j, F_j and H_j; `level` is categorical, naming the type of response variable, with values M, F and H; `year` is categorical, giving the year; `varb` is the interaction of `level` and `year`; and `hdct` is short for `level` $= H$.

We fitted many models; see Appendix D of the technical report at http://www.stat.umn.edu/geyer/aster for details. Scientific interest focuses on the model comparison shown in Table 1.

The models are nested. The affine predictor for Model 4 can be written

$$\varphi_j = \mu_{L_j,Y_j} + \alpha_{L_j} U_j + \beta_{L_j} V_j + \gamma_{L_j,P_j}, \tag{21}$$

where L_j, Y_j, U_j, V_j and P_j are `level`, `year`, `ewloc`, `nsloc` and `pop`, respectively, and the alphas, betas and gammas are regression coefficients. Model 3 is the submodel of Model 4 that imposes the constraint $\gamma_{M,P} = \gamma_{F,P}$, for all populations P. Model 2 is the submodel of Model 3 that imposes the constraint $\gamma_{M,P} = \gamma_{F,P} = 0$, for all P. Model 1 is the submodel of Model 2 that imposes the constraint $\gamma_{L,P} = 0$, for all L and P.

All models contain the graph node effect, `varb` or μ_{L_j,Y_j}, and the quantitative spatial effect, `level:(nsloc + ewloc)` or $\alpha_{L_j} U_j + \beta_{L_j} V_j$, which was chosen by comparing many models; for details see the technical report. We explain here only differences amongst models, which involve only categorical predictors. In an unconditional aster model, which

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these are, such terms require the maximum likelihood estimates of mean-value parameters for each category, summed over all individuals in the category, to match their observed values: 'observed equals expected'. Model 4 makes observed equal expected for total head count $\Sigma_j H_j$, for total flowering $\Sigma_j F_j$, and for total survival $\Sigma_j M_j$ within each population. Model 3 makes observed equal expected for total head count $\Sigma_j H_j$ and for total nonhead count $\Sigma_j (M_j + F_j)$ within each population. Model 2 makes observed equal expected for total head count $\Sigma_j H_j$ within each population.

From purely statistical considerations, Model 3 is the best of these four nested models. Model 4 does not fit significantly better. Model 2 fits significantly worse. It is difficult to interpret Model 3 scientifically, because 'nonhead count' $\Sigma_j (M_j + F_j)$ is unnatural, in effect scoring 0 for dead, 1 for alive without flowers or 2 for alive with flowers.

Model 2 is the model of primary scientific interest. Evolutionary biologists are fundamentally interested in fitness, but it is notoriously difficult to measure; see Beatty (1992), Keller (1992) and Paul (1992). The fitness of an individual may be defined as its lifetime contribution of progeny to the next generation. For these data the most direct surrogate measure for fitness is total head count $\Sigma_j H_j$. The currently available data represent a small fraction of this plant's lifespan. To obtain more complete measures of fitness, we are continuing these experiments and collecting these data for successive years.

Biologists call all our measured variables, the M_j, F_j and H_j, 'components of fitness'. Since M_j and F_j contribute to fitness only through H_j, in an aster model the unconditional expectation of H_j, its unconditional mean-value parameter, completely accounts for the contributions of M_j and F_j. Strictly speaking, this is not quite true, since we do not have H_j measured over the whole life span, so the last M_j contains the information about future reproduction, but it becomes more true as more data are collected in future years. Moreover, we have no data about life span and do not wish to inject subjective opinion about future flower head count into the analysis.

The statistical point of this is that the M_j and F_j are in the model only to produce the correct stochastic structure. If we could directly model the marginal distribution of the H_j, but we cannot, we would not need the other variables. They are 'nuisance variables' that must be in the model but are of no interest in this particular analysis; the mean-value parameters for those variables are nuisance parameters. Model 3 is the best according to the likelihood-ratio test, but does it fit the variables of interest better than Model 2? We do not know of an established methodology for addressing this issue, so we propose looking at confidence intervals for the mean-value parameters for total flower head count shown in Fig. 2.

Although we have no formal test to propose, we claim that it is obvious that Model 3 is no better than Model 2 at 'predicting' the best surrogate of expected fitness. We take this as justification for using Model 2 in scientific discussion and infer from it significant differences among the populations in flower head count and, thus, fitness.

It being difficult to interpret Model 3 scientifically, Model 4 is the next larger readily interpretable model. Model 4 does fit significantly better than Model 2, $p = 0.00016$, which implies differences among populations in mortality and flowering, the M_j and F_j, that may be of scientific interest even though they make no direct contribution to fitness, since Model 2 already fully accounts for their contributions through H_j.

Note that we would have obtained very different results had we used a conditional model, not shown; see §D.3·2 of the technical report. The parameters of interest are unconditional expectations of total flower head count. This alone suggests an unconditional

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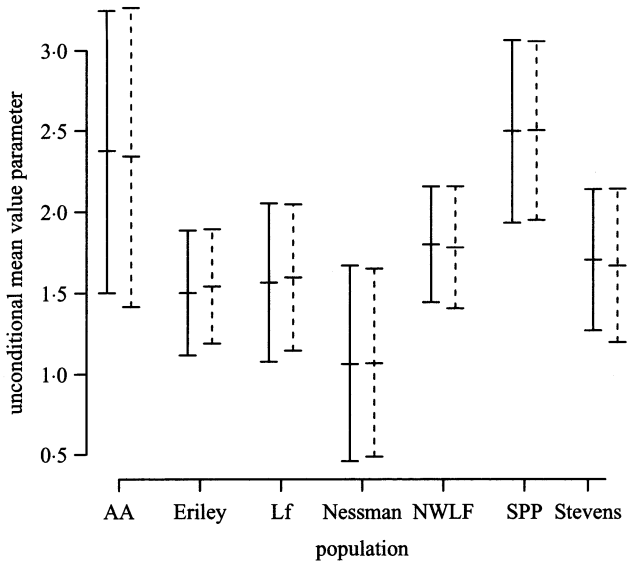


Fig. 2. Confidence intervals for total head count. Shown are 95% nonsimultaneous confidence intervals for the unconditional expectation of total flower head count, all 3 years, for individuals from different populations and central spatial location. Solid bars are based on Model 2 in Table 1, and dashed bars are based on Model 3 in Table 1.

model. Furthermore, we see in (5) that unconditional aster models 'mix levels' passing information up from children to parents. This is why Model 2 in our example was successful in predicting total head count while only modelling `pop` effects at head count nodes. Since a conditional aster model does not mix levels in this way, it must model all levels and so usually needs many more parameters than an unconditional model.

6. DISCUSSION

The key idea of aster models, as we see it, is the usefulness of what we have called unconditional aster models, which have low-dimensional sufficient statistics (9). Following Geyer & Thompson (1992), who argued in favour of exponential family models with sufficient statistics chosen to be scientifically interpretable, an idea they attributed to Jaynes (1978), we emphasize the value of these models in analyses of life histories and overall fitness.

We do not insist, however, the R package `aster` is even-handed with respect to conditional and unconditional models and conditional and unconditional parameters. Users may use whatever seems best to them. Any joint analysis is better than any separate analyses of different variables. We have, however, one warning. In an unconditional aster model, as in all exponential family models, the map from canonical parameters to mean-value parameters is monotone; with sufficient statistic vector Y given by (9) having components Y_k we have

$$-\frac{\partial^2 l(\beta)}{\partial \beta_k^2} = \frac{\partial E_\beta(Y_k)}{\partial \beta_k} > 0. \tag{22}$$

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This gives regression coefficients their simple interpretation: an increase in β_k causes an increase in $E_\beta(Y_k)$, other betas being held constant. The analogue of (22) for a conditional model is

$$-\frac{\partial^2 l(\beta)}{\partial \beta_k^2} = \frac{\partial}{\partial \beta_k} \sum_{G \in \mathcal{G}} \sum_{j \in G} E_\beta(X_j | X_{p(G)}) m_{jk} > 0, \tag{23}$$

where m_{jk} are components of M. Since we cannot move the sums in (23) inside the conditional expectation, there is no corresponding simple interpretation of regression coefficients. Conditional aster models are therefore algebraically simple but statistically complicated. Unconditional aster models are algebraically complicated but statistically simple. They can be simply explained as flat exponential families having the desired sufficient statistics. They are the aster models that behave according to the intuitions derived from linear and generalized linear models.

We saw in our example that aster models allowed us to model fitness successfully. In medical applications, Darwinian fitness is rarely interesting, but aster models may allow data on mortality or survival to be analysed in combination with other data, such as quality of life measures.

ACKNOWLEDGEMENT

Our key equation (5) was derived in about 1980 when R. G. S. was a graduate student and C. J. G. not yet a graduate student. R. G. S.'s Ph. D. adviser Janis Antonovics provided some funding for development, from a now forgotten source, but which we thank nevertheless. High quality life history data, generated recently by S. W., Julie Etterson, David Heiser and Stacey Halpern, and work of Helen Hangelbroek, supported by the U. S. National Science Foundation, about what in hindsight we would call a conditional aster model analysis of mortality data, motivated implementation of these ideas. Were it not for Janis Antonovics' enthusiastic encouragement at that time, we might not have been able to do this now. We should also acknowledge the R team. Without R we would never have made software as usable and powerful as the `aster` package. We also thank Robert Gentleman for pointing out the medical implications.

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[*Received September* 2005. *Revised September* 2006]

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